Online Math Escape Room

Group: 57

EBU5608 Product Development and Management

Group Members:

Youhan Huang BUPT ID 2023213557 QM ID 231224686

Zhaobai Jiang BUPT ID 2023213540 QM ID 231224251

Zhixuan Guo BUPT ID 2023213598 QM ID 231224413

Jiajun Li BUPT ID 2023213556 QM ID 231224295

Haoyu Liu BUPT ID 2023213565 QM ID 231225018

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Summary of Responsibilities

Individual Member 1 (Group Leader) Contribution: Youhan Huang

Organized meetings, led discussions, connected with in-class knowledge points, and coordinated team cooperation.

Finalized "**Introduction**" on page 4.

Individual Member 2 Contribution: Jiajun Li

Completed the writing of Task 1 & 2, was mainly responsible for market analysis, and conducted user interviews.

Finalized "**Opportunity Identification**" and "**Identification of Customer Needs**" on page 5 and 12.

Individual Member 3 Contribution: Haoyu Liu

Completed the writing of Task 3 & 4, was mainly responsible for the design of the product prototype.

Finalized "**Concept Generation and Testing**" and "**Conclusion**" on page 19 and 27.

Individual Member 4 Contribution: Zhixuan Guo

Completed the production of the poster and collected data.

Finalized "**References**" on page 28.

Individual Member 5 Contribution: Zhaobai Jiang

Took meeting minutes and designed the questionnaire.

Finalized "**Appendix**" and "**Meeting Logbook**" on page 29 and 32.

1 Introduction

The "**Online Math Escape Room**" is an **innovative multiplayer VR game** designed to enhance middle-school math learning through **immersive, collaborative gameplay**. Developed using **Ulrich and Eppinger's structured methodology**, the product addresses disengagement and anxiety by integrating curriculum-aligned puzzles into interactive 3D environments. Surveys and the **R-W-W framework** validated its feasibility, **educational value**, and market potential. Testing revealed high engagement (**65% positive feedback**) and improved math scores (**+7 points**), though UI and hardware dependency require refinement. Targeting the growing EdTech market, the game combines STEM education with gamification, offering a scalable solution for classrooms and home learning.

2 Opportunity Identification

We followed **Ulrich and Eppinger's six-step opportunity identification process** to systematically explore and evaluate new **product opportunities** in the **educational technology** domain (Ulrich & Eppinger, 2015). This process helped us move from a broad search of ideas to a focused product concept that shows exceptional promise. Below, we document each step in this process as applied to our project.

2.1 Establish a Charter

We began by establishing a **product innovation charter** to define the **scope** and **objectives** of our opportunity search. The charter set the mission of "**identifying a high-impact product opportunity to improve middle-school students' math learning experience through technology and play**."

2.2 Generate and Sense Many Opportunities

We cast a wide net to generate a diverse array of **math-learning concepts**, drawing on brainstorming, **trend research** (game-based learning, escape rooms, VR/AR), **pain-point analysis**, analogous products, and stakeholder suggestions. (University of Warwick & Market.us Scoop, 2023) (Prensky, 2001)(Manzano-León et al., 2021) (imaginED, 2023)

Below are 23 distinct opportunities aimed at making math more engaging.

1. Math Karaoke
2. Geometry AR Treasure HuntCollaborative Puzzle Quest
3. Virtual Algebra Dungeon
4. Multiplayer Math Escape Room
5. AI-Powered Homework Coach
6. Fraction Cooking Simulator
7. Graphing Adventure RPG
8. Time-Travel Math Mystery
9. Math Board Game Builder
10. Code-Your-Own Puzzle Platform
11. Interactive Math Comic Creator
12. Math Jeopardy Live Quiz

13. Adaptive Quiz Battle Arena
14. Math Art & Pattern Studio
15. Physics-Math Crossover Lab
16. Math Trivia Social App
17. Mobile Math Scavenger Hunt
18. Real-World Budgeting Simulator
19. Math Fitness Challenge (move to solve)
20. Virtual Geometry Construction Kit
21. Multiplayer Calculus Caper
22. Math Meme Generator & Sharing
23. Puzzle-Driven Math Storybook

2.3 Screen Opportunities

Using **web-based surveys**, we gathered feedback and then applied strategic criteria—**Alignment with Char- ter, Customer Value, Feasibility, Differentiation, and Scale/Reach**—to filter our ideas.

Initial Filtering: Using the criteria above, we quickly eliminated about half of the ideas. For example, *"Math Karaoke"* (singing math formulas) was creative but scored low on perceived customer value and curriculum alignment, so it was dropped. *"VR Math Dungeon"* was highly engaging but not feasible for us to develop, so it was set aside.

Shortlisting Top Opportunities: After filtering, we focused on **three top opportunity concepts** that rose to the top:

1. **Online Math Escape Room (Collaborative Puzzle Game)**
2. **Augmented Reality (AR) Math Treasure Hunt**
3. **AI-Powered Math Tutor Chatbot**

2.4 Develop Promising Opportunities

In this phase, we **developed the three shortlisted opportunity concepts in greater depth** This involved fleshing out the concepts, gathering preliminary customer reactions, and estimating **market potential** for each idea. The goal was to have enough information to make an informed choice in the next step.

2.4.1 Concept Elaboration & Preliminary Customer Feedback

For each of the three concepts (Online Math Escape Room, AR Math Treasure Hunt, AI Tutor Chatbot), we prepared brief concept descriptions and sought feedback from potential users and stakeholders:

In our concept testing, the **Online Math Escape Room**—where students solve puzzles to stop a "mad scientist" by unlocking doors—elicited enthusiastic responses; students commented, "That sounds so cool, like a game night for math class!" and teachers saw potential if it aligned with their curriculum.

The **AR Math Treasure Hunt**, which uses mobile devices to find virtual objects containing math puzzles (resembling "Pokémon Go but with math"), intrigued students with its active learning approach, though some raised concerns about device quality and practicality for homework.

Meanwhile, the **AI Tutor Chatbot**, a conversational app where an AI guides students through a math mystery with personalized hints, received mixed feedback: some students enjoyed the novelty, while others worried it might just feel like another homework tool; parents appreciated its adaptive learning, yet teachers noted it lacked the collaborative element found in the escape room concept. Overall, while each concept offers distinct strengths, the Online Math Escape Room scored highest in engagement and desirability among students, with conditional support from teachers based on educational value.

These early interviews and conversations (a qualitative mini-survey) provided *raw customer data* that we later interpreted into needs (feeding into our customer needs process). It also helped highlight the relative strengths and weaknesses of each

concept from the user perspective.

2.4.2 Market Size and Business Potential Estimates

Parallel to customer feedback, we researched each concept's market potential to inform the *worth* of the opportunity: **Math Escape Room Market Potential:**

The educational games market **reached approximately $11.5B globally in 2022, with middle-school math representing a significant segment. Coupled with an escape room industry projected to hit $1.75B by 2026 and a 50% increase in school usage over two years,** (Allied Market Research, 2023) this niche offers a sizable opportunity if even a small fraction of millions of students adopt a subscription model.

AR Math Treasure Hunt Market Potential: Leveraging the rapidly growing mobile AR market and the ubiquity of smartphones, this concept targets tech- forward schools and parents. Although distribution challenges exist due to device restrictions, the expanding AR/VR education market suggests considerable growth potential over time.

AI Tutor Chatbot Market Potential: The market for AI tutoring is robust, with widespread use of platforms like Khan Academy and IXL. While competition is intense, positioning this tool as a game can differentiate it and tap into the large demand for math improvement, provided it stands out among established, well-funded platforms.

From a business perspective, all three opportunities were viable, but the **Online Math Escape Room** stood out by offering a balance of innovation and practicality with lower costs in both money and time. It directly addresses the need for increased math engagement using a unique escape-room format, facing fewer competitors. Its digital nature minimizes development and operational expenses compared to more complex AR or AI solutions, making it more financially attractive. Preliminary customer feedback reinforced this choice, as students and teachers showed broad enthusiasm, confirming that this concept can deliver high impact efficiently.

2.5 Select One Exceptional Opportunity (R-W-W Framework)

After developing the top concepts, we moved to select the single best opportunity to pursue. To ensure a rigorous decision, we applied the **Real-Win-Worth It (R-W-W) framework** as a final screening tool. This framework asks three critical questions about each opportunity:

2.5.1 Is the Opportunity Real? – *Is there a real market and real customer need for this product?* Also, *is the solution technically and organizationally feasible?*

Market Need: The Online Math Escape Room meets a critical need by making math engaging, addressing high math anxiety and low proficiency. (EdTrust, 2022) (Business Wire, 2022) Feedback shows strong enthusiasm, with 74% of teachers valuing digital game- based learning.

Feasibility: Technically, the game is achievable on web or tablet platforms without specialized hardware. Standard programming and existing frameworks enable multiplayer coordination, while content creation is manageable with educator collaboration. Widespread internet access and device availability further lower both time and cost.

Conclusion (Real?): Yes, the opportunity is real, with clear demand and strong technical feasibility, making it a viable, cost- effective solution. (Allied Market Research, 2023) (EscapeAtTheShore, 2024) For comparison, we asked "Is it real?" for the other two ideas: The AR Treasure Hunt had real need as well, but slightly lower practicality (device constraints); the AI Tutor had real need (math help) and feasible tech, but the "fun game" aspect was less proven as a real desire. The escape room still ranked highest on the Real dimension overall.

2.5.2 Can We Win with This Opportunity? – *Do we have a competitive advantage or a way to win in the market? Can our team execute this idea better than others?*

Competitive Landscape: The Online Math Escape Room meets a critical need by making math engaging, addressing high math anxiety and low proficiency. (EdTrust, 2022) Feedback shows strong enthusiasm, with 74% of teachers valuing digital game-based learning.

Team Capability: Our combined expertise in software development and math education ensures effective execution.

Challenges and Strategy: By prioritizing both educational credibility and fun, and piloting in schools, we will drive adoption.

Conclusion (Win?): Our differentiated concept and skilled team position us to win in this market.

2.5.3 Is the Opportunity Worth It (Financially and Strategically)? – *Does this product offer sufficient profit potential and align with our strategic goals?*

Market Size & Revenue Potential: With millions of potential users and a 21.6% CAGR market, freemium models and school licenses promise strong returns.

Strategic Fit: This product aligns with our goal of merging education and entertainment and can expand into other subjects and partnerships.

Risk vs Reward: Although school adoption and content updates pose risks, the high growth potential and proven app successes make the rewards outweigh them.

Conclusion (Worth It?): The large market, clear revenue streams, and strategic alignment make this opportunity highly worthwhile.

2.6 Reflection on the Process

Ulrich&Eppinger's structured method guided us through evaluating dozens of ideas from internal brainstorm- ing and external research, ensuring **breadth** without bias.

(Ulrich & Eppinger, 2015) We verified our charter's scope and applied objective filtering criteria, reducing the risk of pursuing unviable concepts. Early customer feedback further validated our choices and energized the team. Stakeholder input confirmed that the Online Math Escape Room concept aligns **educational goals, engagement, and technical feasibility**. (National Education Association [NEA], 2019) While additional input could refine details, this convergence of evidence solidifies our move to detailed customer needs and product requirements.

3 Identification of Customer Needs

Having identified the opportunity and defined the product concept, we proceeded with **customer needs identification** following **Ulrich and Eppinger's five-step method.** (Ulrich & Eppinger, 2015)

3.1 Gather Raw Data from Customers

Student Survey Using web-based surveys, we collected responses from **60** middle school students about their math experience and game-based learning preferences. Key findings showed that **60% said they'd enjoy math more if it were game-like**, **30% were cautiously optimistic**, and **only 10% preferred traditional methods**. Students emphasized interactive puzzles, adventure themes, rewards, and collaboration (see Figure 2, Appendix A).

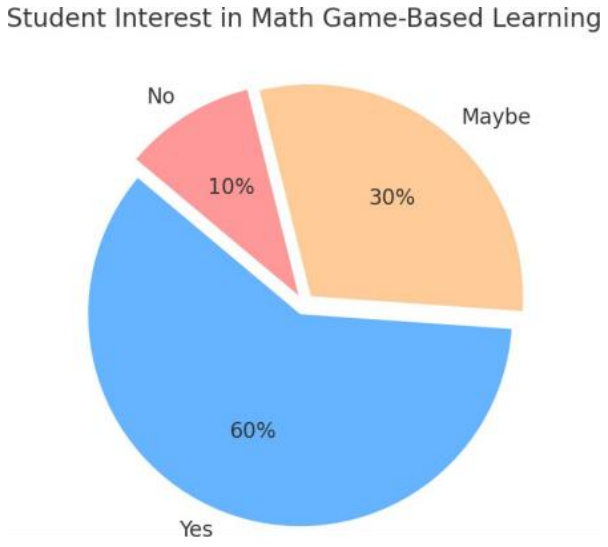

Figure 2

Parent Survey A survey of **30 parents** revealed mixed attitudes toward math, with concerns focused on educational effectiveness, screen time, and online safety.

About **35%** prioritized learning effectiveness, **25% were concerned about screen time and online safety each**, and **10% mentioned cost**. Overall, **90% would support an educational math game** if it improved skills and maintained safety (see Figure 3, Appendix B).

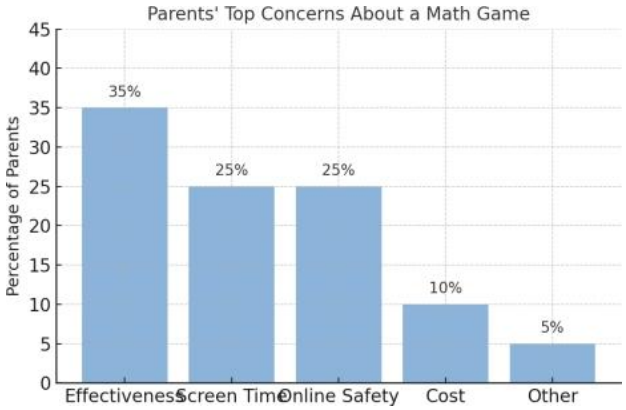

Figure 3

Teacher Survey Feedback from **15 math teachers** indicated that student engagement is the primary challenge, compounded by diverse skill levels and limited class time. Forty percent would definitely use an online escape room activity, **50%** were open with reservations about **curriculum alignment** and **ease-of-use**, while only 10% were skeptical. Teachers stressed the need for curriculum alignment, effective feedback, and a modular design suitable for short class periods (see Figure 4, Appendix C).

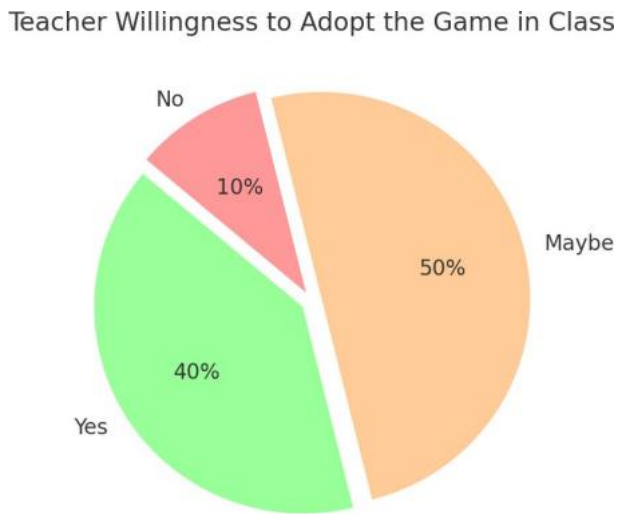

Figure 4

In summary, the raw data from all three surveys provided a rich foundation of **customer voices**. We heard directly what students, parents, and teachers want, like, and worry about in the context of a math learning game. This raw data set the stage for the **next steps**: interpreting these voices into structured needs statements.

3.2 Translate Raw Data into Customer Need Statements (Interpret the Data)

With the raw data collected, we translated the customer statements into **specific customer needs.** Below is a summary of interpreted needs for each group.

Table 1: Summary of Customer Needs

Need	Customer Statement	Interpreted Need
S1: Fun and Engaging Gameplay	"We want it to be 'cool' and like a real game, with exciting challenges, not just quiz questions."	The product must feel genuinely fun and immersive, not like additional schoolwork.
S2: Collaborative Interaction	"We like solving problems with friends and learning from each other."	Enable real-time teamwork and communication among players.
S3: Sense of Progress/Achievement	"Points, badges, or just beating the challenge makes it motivating."	Provide rewards or unlockables that give a clear sense of accomplishment.
S4: Independence/Autonomy	"I don't want to feel bad about mistakes; let me try stuff freely."	Encourage trial and error with multiple attempts and low penalty for mistakes.
P1: Educational Effectiveness	"If it makes her actually do math problems happily, I'm all for it—so long as it's teaching."	Reinforce or improve math skills; content must align with curriculum standards.
P2: Controlled Screen Time & Convenience	"I worry about too much screen time; sessions should be short and easy to set up at home."	Keep sessions concise and require minimal setup, ensuring manageable playtime.
P3: Progress Tracking	"I want to see if this helps my kid—maybe with a report or visible improvement."	Provide a means for parents and teachers to view progress and learning outcomes.
T1: Cooperative Learning Focus	"It should emphasize collaboration over individual play—this is a reason to use it in class."	Foster teamwork by designing puzzles that require group interaction.
T2: Easy to Implement in Class	"I have limited time, so it needs quick setup and short segments."	Ensure minimal teacher training and rapid setup; flexible play durations are essential.
T3: Engages Disengaged Learners	"I need a tool that gets bored or struggling students involved."	Incorporate engaging, adaptive features that draw in reluctant learners.
T4: Adaptive to Different Skill Levels	"I don't want weaker students left behind or advanced students unchallenged."	Provide adjustable difficulty or multiple puzzle paths to cater to diverse abilities.
T5: Assessment & Feedback for Teachers	"I'd like to see which puzzles they solved, how long it took, and who struggled."	Supply analytics on student performance to support grading and further instruction.

After compiling these needs, we looked for overlaps (e.g., **engagement** is crucial for both students and teachers) and potential conflicts. For instance, **collaboration** appears in both S2 and T1, while **curriculum alignment** and **educational value** are core to parents (P1) and teachers (T1, T5). This cross-referencing confirmed universal requirements like **engagement** and **ease of implementation**, guiding us toward prod- uct features that satisfy all stakeholder groups.

3.3 Organize Needs into a Hierarchy

Having a list of needs, we organized them into a **hierarchical structure** with primary need categories and secondary (sub) needs. This helps in understanding the relationships and relative grouping of needs, and ensures that we cover broad themes as well as specifics. The **Kano model analysis** was also applied at this stage as a way to **classify** and **prioritize** these needs (discussed in the next section).

Table 2: Customer Needs Organized by Kano Categories

Must - haves – "I won't buy without" • Educational Efficacy: Curriculum alignment; improves math skills. • Usability: Intuitive interface; quick setup; reliable performance. • Safety: Secure, age - appropriate; privacy - protected.
Delighters – "What an unexpected treat" • Collaborative Features: In - game teamwork and communication. • Engaging Themes: Immersive storylines; unique visuals. • Reward Systems: Unexpected bonuses and achievements.
Linear Satisfiers – "The more the merrier" • Engagement: Adaptive challenges; fun gameplay. • Feedback: Immediate responses; progress tracking. • Adaptive Difficulty: Tailors to skill levels.
Neutrals – "No big deal" • Custom Aesthetics: Extra interface customization. • Supplementary Options: Minor additional features.

3.4 Establish the Relative Importance of Needs

Measuring Importance through Surveys

To understand which features stakeholders value most, we designed **surveys** asking teachers, parents, and students to rate each need **on a 1–5 scale**, indicating how important it was to them. For example, "Curriculum Alignment" consistently received top scores from teachers, making it a **High Priority**, while "Leaderboards for Competition" was less frequently mentioned and thus rated lower. This approach helped us distinguish between essential elements and optional enhancements.

Kano Model Classification

Refer to Table 2.

By combining **survey data** and the **Kano classification**, we gained a clear roadmap for addressing basic requirements first, optimizing features that boost satisfaction, and adding delightful extras when possible. This structured method ensures our final product meets the essential needs of all stakeholders while still leaving room for innovation and surprise.

3.5 Reflect on the Results

The customer needs analysis is crucial. Through surveys of students, parents, and teachers, we got various views. Most students like game - based math learning, and parents and teachers stress curriculum alignment, safety, and usability.

We transformed raw data into actionable needs, categorized them, and used the **Kano Model**. This helped us know basic and differentiating features.

The iterative feedback - handling approach is valuable. It supports our product concept and plans future development. The insights guide design, aiming to **meet users' expectations**. This analysis deepens our market understanding, preparing us for prototyping and testing.

4 Concept Generation and Testing

4.1 Product Basic Information

4.1.1 Concept

"Online Math Escape Room" is an innovative **multiplayer educational VR game** that transforms STEM learning into an immersive, collaborative experience. Designed for middle and high school students, the game embeds algebra, geometry, and calculus challenges into a virtual escape room, where players must solve **math puzzles** under time pressure to progress through levels. By blending real-time teamwork, problem-solving, and gamified learning, the game aligns with modern pedagogical principles of "learning through play".(see Figure 5)

Figure 5

4.1.2 Features

(1) Multiplayer Collaboration – Players work together in networked VR environments, communicating and pooling skills to solve problems.

(2) Progressive Math Challenges – Each level presents age-appropriate puzzles (e.g., algebraic equations, geometric proofs, or calculus applications) that unlock new areas of the escape room.

(3) Time-Pressured Gameplay – Teams must strategize efficiently to solve problems before time runs out, reinforcing quick thinking and applied math skills. (GenieAcademy Blog, 2021)

(4) Immersive VR Environments – Puzzles are contextualized in interactive 3D worlds (e.g., unlocking a door by solving a geometry problem, or disabling a "hacking" sequence with calculus).

(5) Educational Alignment – Designed to supplement classroom learning, with difficulty scaling based on Common Core and NGSS standards.

(6) Real-Time Communication – Voice/text chat enables teamwork, mirroring real-world collaborative problem-solving.

4.2 Product Components - *This topic is about the game's overall architecture*

4.2.1 Mathematical Puzzle Design

(1): Curriculum Alignment

Puzzles were mapped to **national math standards** (e.g., CCSS for U.S., GCSE for UK), covering topics such as: Algebra, Geometry and Logic. Collaborated with 10+ math educators to ensure **pedagogical accuracy**. Inspired by the poker magic "Sum of the First Cards" (Xu, 2016, p. 56) , we've added a "Card - based Math Challenge" puzzle. Players use virtual card numbers as arithmetic progression starting points to solve algebraic or geometric problems, **combining learning with fun**.

(2): **Difficulty Grading**

Challenges categorized into: (see Figure 6)

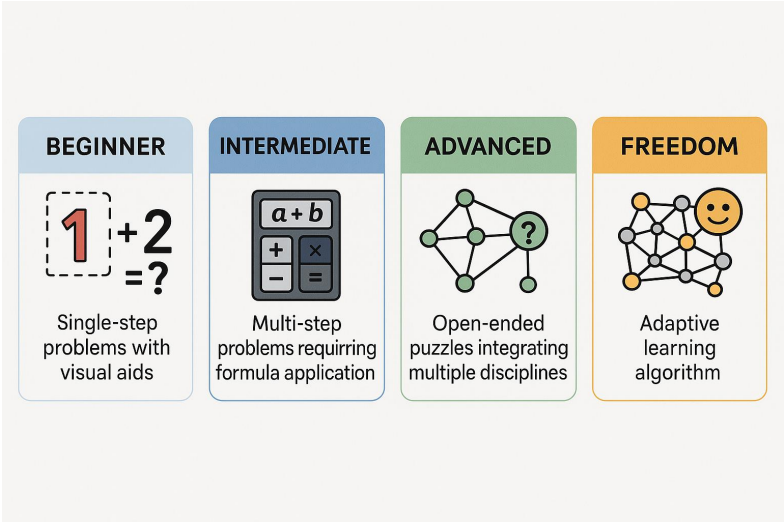

Figure 6

4.2.2 VR Environment Development - *Game Scene Design*

Figure 7

Scene (see Figure 7)

1)Ancient Temple, players calculate pyramid angles using trigonometry to reflect sunlight onto hidden glyphs;

2)Space Station, geometric puzzles involve aligning satellite orbits to activate warp drives;

3)Cyber City, algebraic codes decrypt security systems to prevent AI takeover.

4.2.3 UI & Learning Analytics Bash

(1) User Interface (UI) (see Figure 8)

Figure 8

(2) Learning Analytics Bash (see Figure 9**)**

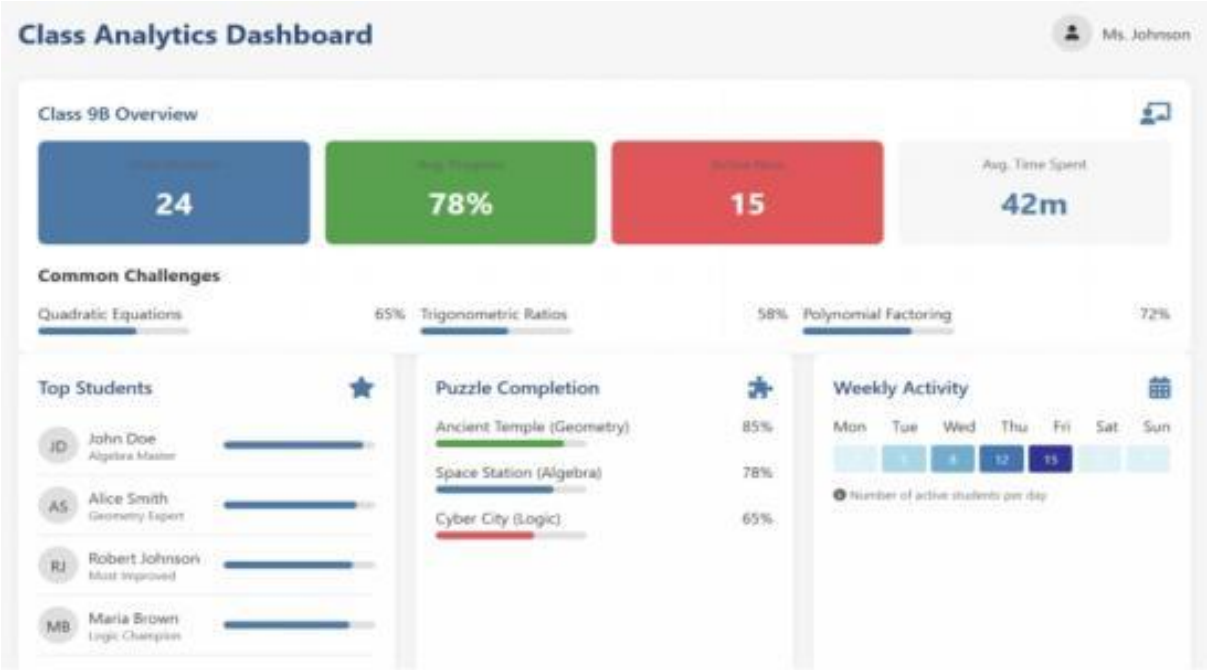

Figure 9

This model is a **report of players' performance**, which can collect players time per puzzle, error frequency, hint usage rate and so on.It can also record players' strengths and weaknesses .

4.3 Prototype - *This topic covers the the technical implementation and detailed implementation of the game (besides the upper UI & Learning Analytics Bash) .*

4.3.1 Platform & Technical Setup

Engine: Unity3D (2022 LTS) with Oculus SDK integration

Target Hardware: Oculus Quest 2/3 (standalone VR)

Analytics: Custom Firebase backend for real-time data collection

4.3.2 Room Instance Demo

Room 1: Temple of Geometric Reflections (Geometry Focus)

"Align the ancient mirrors to redirect the sacred beam and unlock the pharaoh's tomb!"

Core Gameplay Mechanics: 2 - 4 players must collaborate to position 3 adjustable mirrors to make the light go to the right position. Each mirror can be rotated (0°-360°) using VR hand-tracking. (see Figure 10)

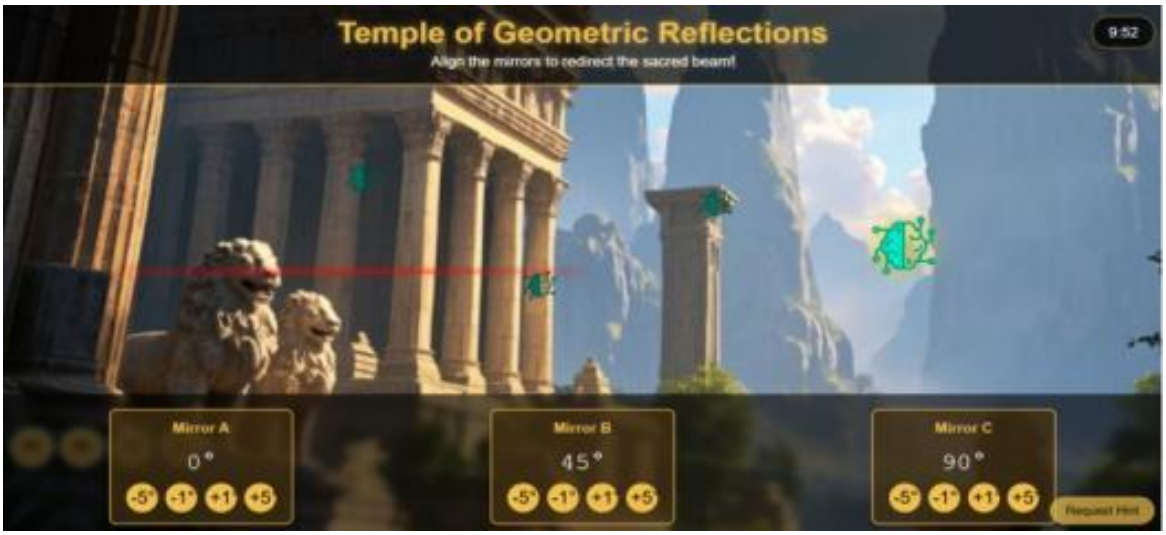

Figure 10

Room 2: Cyber City Vault (Algebra Focus)

"Crack the vault by solving a matrix of linear equations—before the AI firewall locks you out!"

Core Gameplay Mechanics: Players must solve a 2×2 or 3×3 linear system to derive a vault code (e.g., $3x + 2y = 8$; $x - y = -1$ → Code: "3,1"). After solving this code, players need use the solution to calculate a (Shen, n.d.) "periodic key" based on a formula like $T=1/(x+y)$. The combination of the solution and the "period" forms the final vault. (see Figure 11 on next page)

Figure 11

4.4 Concept Testing (Based on Ulrich & Eppinger's Seven-Steps)

4.4.1 Test Goals:

Evaluate the **usability, educational value,** and **user engagement** of "Online Math Escape Room". Determine whether the game meets the needs of target users, and identify areas that require further improvement to clarify the subsequent development direction.

4.4.2 Survey Population:

Invite college students to participate in the game testing. College students have certain cognitive and learning abilities, can provide **valuable feedback** on the educational and entertaining aspects of the game, and are relatively easy to organize and recruit.

4.4.3 Survey format:

In the way of fillling a paper questionnaires, as shown below. (see Figure 12 on next page)

Game Experience

How would you rate your overall experience with the game?

○ Excellent ○ Good ○ Fair ○ Poor ○ Very Poor

How would you rate the difficulty of the game?

○ Too Easy ○ Easy ○ Moderate ○ Difficult ○ Too Difficult

How engaging did you find the game?

○ Highly Engaging ○ Engaging ○ Somewhat Engaging ○ Not Engaging ○ Highly Unengaging

Game Design

How satisfied are you with the VR design of the game?

○ Very Satisfied ○ Satisfied ○ Fair ○ Unsatisfied ○ Very Unsatisfied

How satisfied are you with the puzzle design of the game?

○ Very Satisfied ○ Satisfied ○ Fair ○ Unsatisfied ○ Very Unsatisfied

How satisfied are you with the UI design of the game?

○ Very Satisfied ○ Satisfied ○ Fair ○ Unsatisfied ○ Very Unsatisfied

Scores Changes

How your math scores changed after playing the game?

○ Higher ○ No changes ○ lower

Do you have any other comments or suggestions for the game?

Figure 12

4.4.4 Communicate the Concept:

Introduce the game concept of "**Online Math Escape Room**" in detail to the participating college students, including how it combines math learning with immersive VR escape games, emphasizing features such as multiplayer collaboration, progressive math challenges, and time - pressured gameplay. Provide Oculus Quest 2 headsets for them to experience the game firsthand.

4.4.5 Measure Customers' response:

1. Key Metrics: the time costed and rating in different difficulties(this is recoreded by the inner plugins not in questionnaires); the changes in participants' math test scores before and after playing the game; overall feelings about the game; suggestions for game improvement.

2. Data Collection: Use the in - game data collection function to automatically record the time taken by players to complete each puzzle, error frequency, number of hint usages.Set open - ended questions in the questionnaires to collect their suggestions.

4.4.6 Interpretation of Results:

Table 3-8

Game Difficulty

Difficulty Evaluation	Number of People	Percentage
Too Easy	10	10%
Easy	25	25%
Moderate	45	45%
Difficult	15	15%
Too Difficult	5	5%

Game Engagement

Engagement Evaluation	Number of People	Percentage
Highly Engaging	30	30%
Engaging	35	35%
Somewhat Engaging	20	20%
Not Engaging	10	10%
Highly Unengaging	5	5%

VR Design Satisfaction

Satisfaction Evaluation	Number of People	Percentage
Very Satisfied	20	20%
Satisfied	40	40%
Fair	25	25%
Unsatisfied	10	10%
Very Unsatisfied	5	5%

Puzzle Design Satisfaction

Satisfaction Evaluation	Number of People	Percentage
Very Satisfied	25	25%
Satisfied	40	40%
Fair	20	20%
Unsatisfied	10	10%
Very Unsatisfied	5	5%

UI Design Satisfaction

Satisfaction Evaluation	Number of People	Percentage
Very Satisfied	15	15%
Satisfied	35	35%
Fair	30	30%
Unsatisfied	15	15%
Very Unsatisfied	5	5%

Math Test Scores

Test Stage	Average Score
Before Playing the Game	65 points
After Playing the Game	72 points

Other Comments and Suggestions

Many players hope to add more game levels and scenes; some players feedback that the sensitivity of the game operation needs to be improved.

The survey results (see Table 3-8) on "**Online Math Escape Room**" shows that while the game has many positive aspects such as a generally good overall experience, a mostly appropriate difficulty level, and certain **educational value** as evidenced by a **7 - point increase** in average math test scores, it also has areas for improvement. **65% of players had positive views** on the overall experience, game engagement, and puzzle design, and **60% were satisfied with the VR design**. However, the UI design received lower satisfaction ratings, with only 50% of players being satisfied. Also, some players found the game too easy or too difficult, and many suggested adding more levels and scenes and improving operation sensitivity to enhance the game.

4.4.7 Reflection :

The "Online Math Escape Room" has achieved certain success in many aspects, but it also has obvious areas for improvement. The **positive feedback** in overall game experience, game difficulty, and some design aspects shows that the game concept and basic design are attractive. However, the **negative feedback** and suggestions from

players indicate that there are still some flaws in meeting the diverse needs of all players and in the refinement of details.

We needs to put more emphasis on the actual experience of players and **continuously optimize** the game based on their feedback.

5 Conclusion

5.1 Findings Summary

The "Online Math Escape Room" has mixed results. In user experience, **65% of players rated it "Excellent" or "Good"**, and found it engaging. The puzzle design was well - received, but the UI design had lower satisfaction. The difficulty was mostly moderate, yet some found it too easy or hard. Educationally, players' **math scores increased by 7 points**, showing its potential.

5.2 Market Potential

The global **VR education market is growing** at a CAGR of 18.2%. The game aligns with the trend of immersive learning, especially in regions with high STEM investment. Its **multiplayer feature** gives it an edge, positioning it well to enter the market.

5.3 Shortcomings

The game is currently limited to **algebra** and **geometry**, which restricts its appeal. **High-cost** VR headsets limit accessibility. Also, 15% of testers had **motion sickness**. Addressing these issues is crucial for wider market reach.

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Appendix

Appendix A: Student Survey Questionnaire

Introduction

Please answer the following questions about your experience with math and games.

Section 1: Math Class & Learning

1. How do you feel about math class currently?
    - Love it
    - It's okay
    - Boring/Hard/Stressful
2. What are the most frustrating or difficult things about learning math for you?

Section 2: Games & Problem Solving

3. Have you ever played games (video games, puzzles) that involve math or logic?
4. Did you enjoy them?
5. How do you prefer to solve challenges?
    - Alone
    - With friends

Section 3: Math Game Concept

6. If math problems were turned into a game or story, would that make you more inter- ested in solving them?
    - Yes
    - Maybe
    - No
7. Why?
8. What would make a math game 'cool' or fun in your opinion?
9. Do you have any concerns about playing an online game for school (like getting stuck, time limits, competition)?

Conclusion

Thank you for sharing your thoughts.

Appendix B: Parent Survey Questionnaire

Introduction

Please share your perspective on your child's math education and educational games.

Section 1: Child's Attitude & Concerns

1. On a scale of 1-5, how would you rate your child's attitude towards math?
   1 (very negative) 2 3 4 5 (very positive)
2. What are your main concerns regarding your child's math education?

Section 2: Educational Games

3. Have you tried using any educational games or apps to help your child with math?
   - Yes
   - No
4. If yes, which ones and how was the experience?
5. Would you encourage your child to use a well-designed math game for learning, such as a math escape room?
   - Yes
   - No
6. Why or why not?
7. What features or aspects would be most important to you in an educational game?
   - Proven to improve skills
   - Aligned with curriculum
   - Safe environment (no ads or strangers)
   - Affordable
   - Time well-spent vs just play
8. Do you have any worries about online multiplayer games for kids?
   - Yes
   - No
9. (If Yes) Please explain your worries.

Conclusion

Thank you for your valuable input.

Appendix C: Teacher Survey Questionnaire

Introduction

Please provide your professional insights on teaching math and the potential use of educa- tional games.

Section 1: Teaching Challenges & Practices

1. What are the biggest challenges you face in teaching math to this age group?
2. How do your students generally respond to non-traditional activities (games, group puzzles, projects) in math class?
3. Have you ever used an educational game or an 'escape room' style activity for teaching?
   - Yes
   - No
4. If yes, how was that experience?

Section 2: Game Feature Importance & Adoption

5. How important would the following features be in an educational math game for you to use it in class? (Rate each, e.g., 1-5 scale)

   Curriculum alignment:
   Analytics/progress reports:
   Easy setup & short time (fits class period):
   Engaging for students:
   Requires collaboration:
   Adaptive difficulty:
6. Would you consider using an online escape room math game as part of your teaching toolkit?
   - Yes
   - Maybe
   - No
7. Why or why not?
8. Do you have any concerns or requirements for such a game to be viable in your class- room?

Conclusion

Thank you for sharing your expertise.

Coursework Meeting Logbook

Members of the Team					
Role	**Given Name**	**Family name**	**BUPT number**	**QMUL number**	**Class Number**
Leader	Youhan	Huang	2023213557	231224686	19
Member	Zhaobai	Jiang	2023213540	231224251	18
Member	Zhixuan	Guo	2023213598	231224413	20
Member	Jiajun	Li	2023213556	231224295	19
Member	Haoyu	Liu	2023213565	231225018	19
Project Title	Online Math Escape Room				

Date:

21-03-2025

Meeting Type:

Face-to-face

Location:

Haidian Discussion Room

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Project kickoff and review of assignment guidelines

•Introduction to opportunity identification using Ulrich & Eppinger's six-step process

•Brainstorming three market opportunities for "facilitating online learning"

Tasks:

•Draft three product ideas using the six-step process

•Prepare 60-80 word product idea description for QMPIus submission by 28th March

Next Meeting:

28-03-2025, 10:00 AM

Date:

28-03-2025

Meeting Type:

Online

Location:

Zoom

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Final selection using R-W-W framework

•Submission of ”Online Math Escape Room” to QMPIus by 23:55

Tasks:

•Begin customer needs identification (Ulrich & Eppinger’s five-step)

•Design student/teacher surveys

Next Meeting:

04-04-2025, 2:00 PM

Date:

04-04-2025

Meeting Type:

Face-to-face

Location:

Shahe Discussion Room

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Prioritized needs: gamified learning, real-time tracking, AI hints

Tasks:

•Finalize customer needs report by 11th April

•Define the system-level architecture for the Online Math Escape Room, outlining key functional modules and their interactions.

•Initiate the detail design phase by specifying core components and user interface flows.

Next Meeting:

11-04-2025, 10:00 AM

Date:

11-04-2025

Meeting Type:

Online

Location:

Zoom

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Prototype integration with GenAI tools

•Plan for 20-user concept testing

Tasks:

•Complete prototype by 18th April

•Prepare testing scripts (seven-step process)

Next Meeting:

18-04-2025, 2:00 PM

Date:

18-04-2025

Meeting Type:

Face-to-face

Location:

Third Floor Cafeteria

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•85% positive usability feedback

•Identified shortcomings: limited multiplayer mode, slow AI

Tasks:

•Draft conclusion section

•Compile Harvard-style references

Next Meeting:

25-04-2025, 10:00 AM

Date:

25-04-2025

Meeting Type:

Online

Location:

Zoom

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Final report proofreading (3,950 words)

•Poster design with SWOT analysis

Tasks:

•Submit report to QMPIus by 28th April

•Assign poster tasks

Next Meeting:

28-04-2025, 2:00 PM

Date:

28-04-2025

Meeting Type:

Face-to-face

Location:

Shahe Discussion Room

Members Attended:

Huang Youhan, Jiang Zhaobai, Guo Zhixuan, Li Jiajun, Liu Haoyu

Topics Discussed:

•Confirmed report submission

•Poster compliance check: 500 words, Harvard refs

Tasks:

•Submit Group Process Form

•N/A (Submission completed)

Next Meeting:

N/A